

Supplementary Material: Internal Validation Evidence for OCR and YOLOv11n-OBB

Scope Note

This supplement reports module-level validation results for the OCR and oriented field-detection components. These results support internal validation claims only. They do not establish held-out-test performance, leakage-free generalization, field-level exact match, full-record exact match, per-class detector robustness, or end-to-end information-extraction accuracy.

Table 1: OCR validation summary for internal crop-level recognition. For each complete OCR run, the table reports the number of validation checkpoints, minimum training CER, minimum validation CER, the checkpoint of the minimum validation CER, and the final validation CER. These values describe OCR on cropped text regions only.

Model	Checkpoints	Min train CER (%)	Min val CER (%)	Best val checkpoint	Final val CER (%)	Final checkpoint
GRU-Custom	144	14.521924	29.713823	136	29.786629	144
GRU-MobileNet	134	4.522116	6.216428	118	6.244398	134
LSTM-Custom	100	4.102232	37.863853	84	37.895679	100
LSTM-MobileNet	157	1.151390	5.124146	141	5.145335	157
Transformer	143	0.809839	2.032055	111	2.045730	143

Table 2: YOLOv11n-OBB internal validation maxima and terminal stability. The table reports the best value and epoch, final epoch value, and last-10-epoch mean, standard deviation, minimum, and maximum. These metrics are aggregate detector-validation metrics and do not quantify held-out-test performance, per-class robustness, downstream OCR, or end-to-end extraction accuracy.

Metric	Best	Best epoch	Final	Last-10 mean	Last-10 SD	Last-10 min	Last-10 max
Precision	0.99499	64	0.98574	0.989150	0.001845	0.98574	0.99122
Recall	0.98371	127	0.98310	0.980740	0.001174	0.97926	0.98310
mAP50	0.99396	120	0.99352	0.993421	0.000111	0.99327	0.99359
mAP50-95	0.88233	131	0.88190	0.881679	0.000297	0.88123	0.88208

Interpretive Note

The OCR validation results provide direct evidence for the OCR comparison under internal crop-level validation. Among the complete OCR runs, the Transformer-based model achieved the lowest validation CER, reaching 2.032055% at checkpoint 111 and ending at 2.045730% at checkpoint 143. This result supports a module-level OCR claim for cropped text regions only. It should not be interpreted as field-level exact-match accuracy, full-record exact-match accuracy, held-out-test performance, or end-to-end extraction accuracy.

The YOLOv11n-OBB validation results support the reported detector maxima: precision = 0.99499, recall = 0.98371, mAP50 = 0.99396, and mAP50-95 = 0.88233. The final 10 epochs show stable mAP values near the reported maxima, with mAP50 mean = 0.993421 and mAP50-95 mean = 0.881679. These values

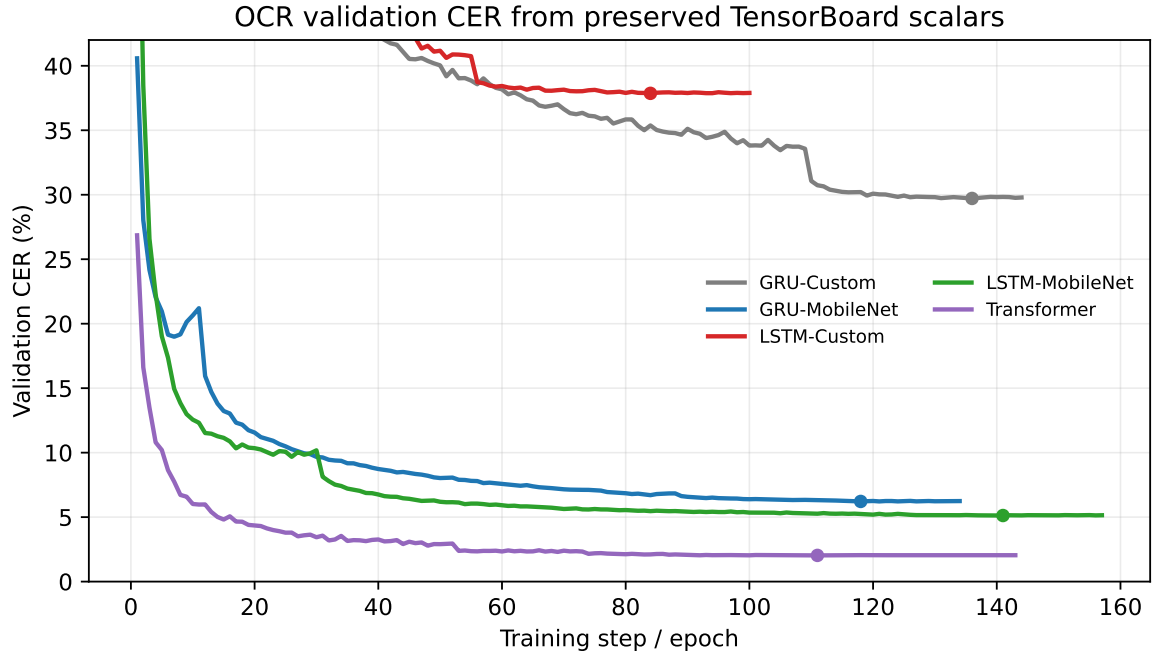


Figure 1: Validation CER over training checkpoints for the complete OCR runs. The Transformer run reaches the lowest internal validation CER among the evaluated models, with a minimum of 2.032055% at checkpoint 111 and a final value of 2.045730% at checkpoint 143. Lower CER indicates fewer character-level edit errors on cropped text regions. The figure should be interpreted strictly as internal crop-level OCR validation; it does not evaluate detector errors, field-level exact match, full-record exact match, or end-to-end extraction accuracy.

characterize the configured detector under internal validation and do not establish independent held-out-test performance, per-class robustness, or end-to-end text-extraction accuracy.

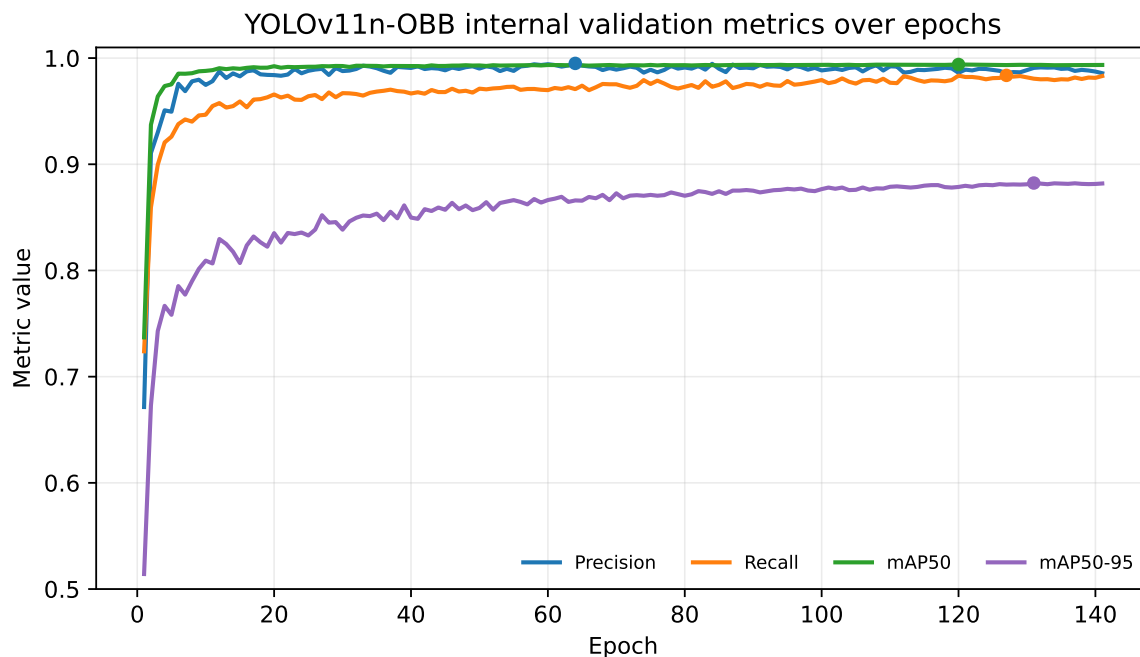


Figure 2: Internal validation trajectories for the configured YOLOv11n-OBb detector over 141 epochs, showing precision, recall, mAP50, and mAP50-95. Reported best values are precision = 0.99499 at epoch 64, recall = 0.98371 at epoch 127, mAP50 = 0.99396 at epoch 120, and mAP50-95 = 0.88233 at epoch 131. These curves summarize detector validation history only and should not be interpreted as independent held-out-test evidence, per-class performance, or end-to-end information-extraction accuracy.

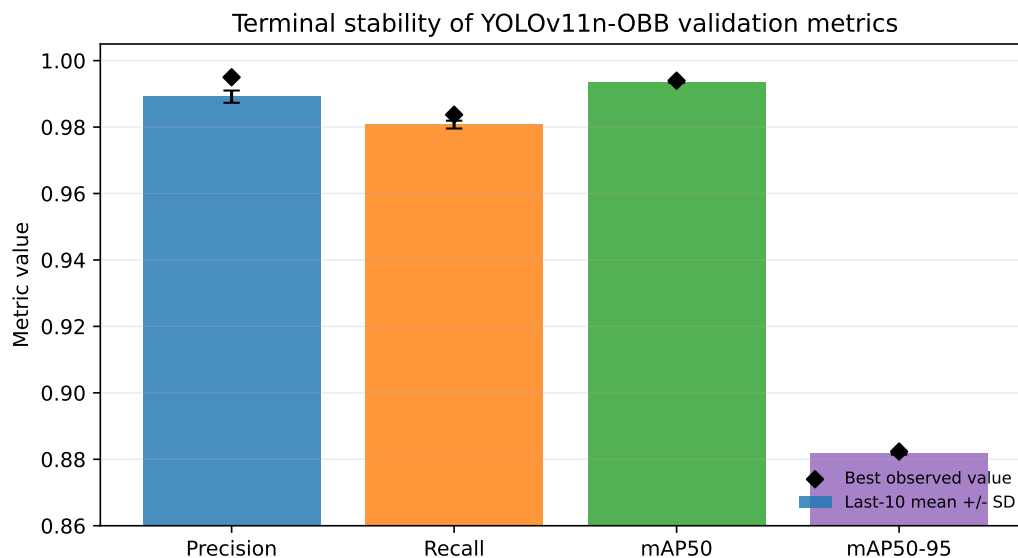


Figure 3: Last-10-epoch stability summary for YOLOv11n-OBb headline validation metrics. mAP50 shows a last-10 mean of 0.993421 with standard deviation 0.000111, and mAP50-95 shows a last-10 mean of 0.881679 with standard deviation 0.000297. This supports that the final detector validation metrics were stable near the reported maxima, but it does not address dataset leakage, external generalization, or downstream OCR correctness.

Table 3: Validation evidence summary for the OCR and YOLOv11n-OBB modules. The table clarifies the scope of each result reported in the supplementary material.

Module or model	Validation scope	Retained result	Interpretation
GRU-Custom	Internal cropped-text validation	Min Val CER 29.713823%	Included in OCR model comparison
GRU-MobileNet	Internal cropped-text validation	Min Val CER 6.216428%	Included in OCR model comparison
LSTM-Custom	Internal cropped-text validation	Min Val CER 37.863853%	Included in OCR model comparison
LSTM-MobileNet	Internal cropped-text validation	Min Val CER 5.124146%	Included in OCR model comparison
Transformer incomplete one-checkpoint run	Internal cropped-text validation	Min Val CER 24.655149%	Excluded from final OCR comparison because the run was incomplete
Transformer complete run	Internal cropped-text validation	Min Val CER 2.032055%	Used as the best OCR result among complete runs
YOLOv11n-OBB	Internal detector validation	141 training epochs	Used for aggregate detector metrics and terminal-stability analysis